



BALL STATE
UNIVERSITY

Module 8 Lecture - Confidence Intervals

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Calculate and interpret confidence intervals for estimating a population mean and a population proportion.
- Interpret the Student's t probability distribution as the sample size changes.
- Discriminate between problems applying the normal and the Student's t distributions.
- Calculate the sample size required to estimate a population mean and a population proportion given a desired confidence level and margin of error.

1.2 Instructor Learning Objectives

- Understand the value of calculating a confidence interval in interpreting the “accuracy” of a certain statistic
- Appreciate how the calculation and interpretation of a confidence interval builds upon our previous understanding of distributions and probability
- Value how confidence intervals demonstrate the inherently probabilistic nature of quantitative analysis in research

1.3 Introduction

! Important

Professional researcher often forget just how important it is to report confidence intervals in results - but they are actually very useful in interpretation!

- Up until now, we have primarily been working with **point estimates** of sample statistics
 - E.g., a statistic of a mean ($\bar{x}$) is a point estimate of the population parameter mean (μ)
 - Same logic applied to standard deviation, variance, median, etc.
- Inherently, our single point estimate of a statistic is going to be insufficient for telling us much about how close it is to the population parameter
 - Thus, we need some other to add to the point estimate to give us more information about the population

! Important

Remember our goal with statistics is to tell us something about the population, not just the sample!

- This is, in essence, getting us closer to **inferential statistics**, where we can actually infer meaning from the characteristics of our _____
- The part that we need to add are called **confidence intervals**
 - A _____ interval is effectively a _____ of values that we believe a population parameter falls in, with a certain amount of confidence
 - It is actually appropriate to refer to the confidence interval itself as a _____ continuous variable, as it is referring to a distribution of possible values that sample statistics can be drawn from

 Discuss: As a review, try explaining, in your own words, what a 'random' variable is?

- At first, we'll talk about applications of confidence intervals to the _____, implying that we are working with continuous variables
 - We'll touch on more categorical, discrete type stuff later

1.4 Nuances in the Confidence Intervals

- Commonly, we use _____ percent as the level of confidence
 - However we can use something like _____ percent for a more _____ estimate

 Discuss: If you have taken statistics before, what other value is commonly associated with 95%

- Also, it is important to treat the confidence interval as another type of estimate, which we usually call a **interval estimate**
 - Just like our _____ estimates, there is no guarantee this is infallible

- In fact, the confidence interval is best understood as a description of the **reliability** of sample statistics when taken from the same population, but does not mean it is certain that the population parameter falls within the range

 Discuss: For review, describe again why we can't take sample estimates as being representative of populations? hint: use the vocabulary sample v —

- Depending on the _____ and the outlet you may see confidence intervals described as something like **margin of error**

1.5 Basic Calculating Confidence Intervals

- In order to _____ the confidence interval of a certain statistic, we need to know the standard _____ of the variable that we are interested in
 - Recall, this is the _____ standard deviation of the distribution that our statistic comes from
- However, this is most easy to know when we can assume that our variable comes from a _____ distribution
 - This comes back to being able to appeal to the _____ rule that was introduced as part of understanding normal distributions
 - Remember that the empirical rule is also sometimes called the 68-95-99.7 rule

 Discuss: Based upon the description above, try re-describing what this rule says about normal distributed variables

! Important

There are several slight variations on the empirical rule, but it is easiest to round things off to whole number standard deviations

- In order to calculate the 95% confidence interval, we use the following formula:
 - $[PE - 2SE, PE + 2SE]$
 - Where PE : is our _____ estimate
 - SE : standard _____ of the statistic
 - $2SE$ is used in appeal to the _____ rule, because, theoretically, 95% of the values of the statistic fall within 2 standard _____; we can use 1 or 3 to find the 68% or 99.7% confidence intervals, respectively

! Important

Recall that the standard error decreases as sample size increases, which also means that a larger sample size results in a more narrow confidence interval

- This becomes practically _____ as 95% of the sample statistics taken from this population with the same sample size (n) would give us a statistic within this range 95% of the time
- For example: I have a point estimate mean statistic ($\bar{x}$) of 100, a sample size (n) of 40, and a population parameter standard deviation (σ) of 10
 - Review: $SE = \frac{\sigma}{\sqrt{n}}$
 - Thus: $SE = \frac{10}{\sqrt{40}} = 1.58$
 - So 95% CIs are: $[100 - 3.16, 100 + 3.16] = [96.84, 103.16]$
 - Practical interpretation: 95% of infinitely many sample statistics taken from this population will fall between 96.84 and 103.16

🔊 Discuss: Try following the same procedures for a point estimate mean of 50, a sample size of 12, and a population standard deviation of 20

2 A Single Population Mean Using the Normal Distribution

2.1 Introduction

- In this section, we'll say more to the _____ and specifics used in each part of the calculation process, when the population standard deviation is _____
 - You may _____ that we can often know the population SD, which is fair, we'll get into that in a bit
 - For this scenario, we rely upon the _____ limit theorem
- We arbitrarily chose our **confidence level (CL)**, but is usually 90% or above; it is how _____ we want to be that the population parameter falls within the confidence intervals
 - Based upon that CL, we can calculate the **error bound for a population mean (EBM)**, which is the amount we deviate (i.e., add or subtract) from the point estimate mean
- Another way of conceptualizing confidence level is through **alpha**
 - It is just the complement of CL: $\alpha + CL = 1$

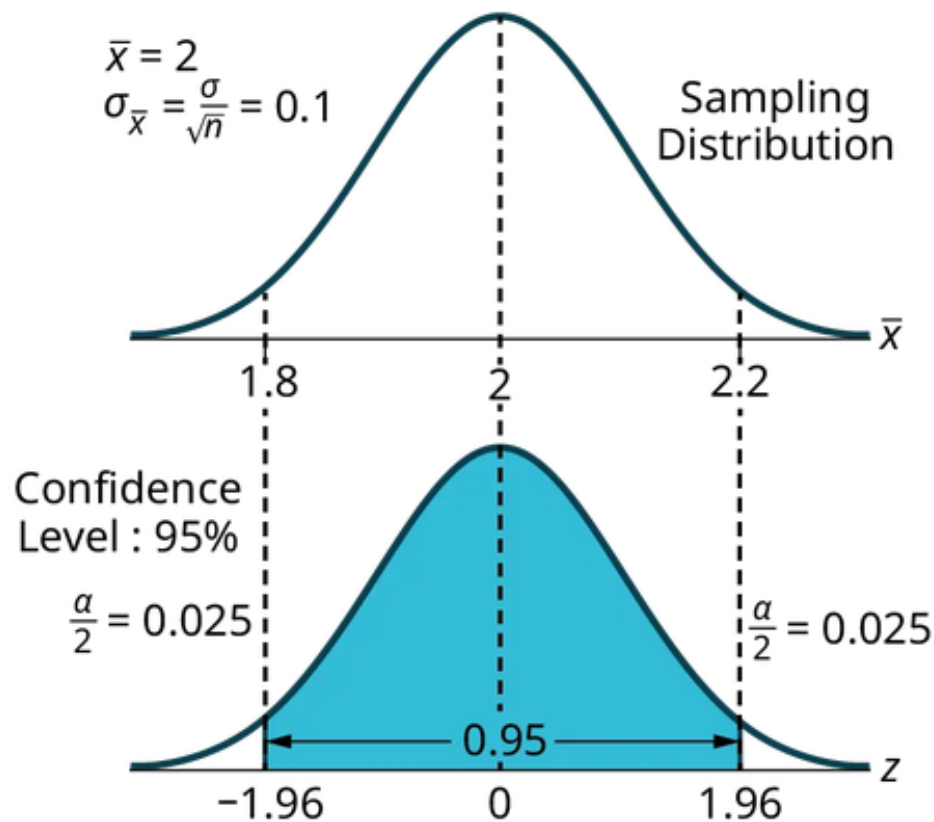


Figure 8.2

$$\begin{aligned}
 \mu &= \bar{X} \pm Z_{\alpha} \left(\frac{\sigma}{\sqrt{n}} \right) \\
 &= 2 \pm 1.96(0.1) \\
 &= 2 \pm 0.196 \\
 1.804 &\leq \mu \leq 2.196
 \end{aligned}$$

! Important

For my folks who haven't taken stats before, alpha will show up again when we talk about p-values!

- Consider my prior example:
 - I have a point estimate mean statistic ($\bar{x}$) of 100, a sample size (n) of 40, and a population parameter standard deviation (σ) of 10
 - * Review: $SE = \frac{\sigma}{\sqrt{n}}$
 - * Thus: $SE = \frac{10}{\sqrt{40}} = 1.58$
 - * So 95% CIs are: $[100 - 3.16, 100 + 3.16] = [96.84, 103.16]$

 Discuss: Identify what is the CL, alpha, and EBM in this worked problem

 Discuss: Make sure you can properly describe the different between a standard deviation of the population and the standard error of the mean

2.2 Writing Z-score Notation for Confidence Intervals

- Thinking back to the image above, we split α between the two _____ of the normal distribution
 - Thus, each tail contains $\frac{\alpha}{2}$ area, and the _____ that denotes where the CI bound is is $z_{\frac{\alpha}{2}}$
 - Therefore the upper 95% CI bound z is stated as $z_{\frac{0.05}{2}}$ or $z_{0.025}$

 Discuss: Now do the same thing with a 92% CI lower bound

3 A Single Population Mean Using the Student t Distribution

3.1 Introduction

- As mentioned prior, we don't often have what we can easily consider to be a population standard deviation
 - In fact, that may be one of the many things we are trying to

 Discuss: Quickly address why you won't regularly have the population parameters and under what circumstances would you have them?

- William Gosset** was staff at the Guinness Brewery and, under the pen name of "Student" came up with the idea of **Student's t-distribution** to be used when the plain normal distribution didn't work
 - Use of Student's t-distribution is now the way to calculate confidence intervals for means

3.2 More on the t-distribution

- steps:
 - Draw a simple sample of size n
 - Population of distribution with parameters mean μ and standard deviation σ
 - Convert of the distribution to t-scores with $\frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$
- Results in a Student's with $n - 1$ **degrees of freedom (df)**
 - The individual t score of a mean is understood as the distance that $\bar{x}$ is from μ , like a
- Degrees of freedom are a tricky concept, but come up in most inferential statistical formulas

- When calculating standard deviation for a sample, we calculate n number of _____ and because we know that the sum of all deviations is 0 (if we don't square them, like in the variance formula), the last deviation cannot vary.
- Thus, all deviations (n) can vary until this _____ one, leading to $n - 1$ as the degrees of freedom in this case

! Important

Perhaps the most defining feature of the t-distribution that separates it from the normal distribution is that it changes with the sample size

- The _____ for the t-distribution is given at $T \sim t_{df}$
 - Where $df = n - 1$, as previously discussed

3.3 Calculating with t-distribution

- In the case that we _____ know the population standard deviation, we can use the t-distribution in the following way
- $EBM = (t_{\frac{\alpha}{2}})(\alpha s \sqrt{n})$
 - $t_{\frac{\alpha}{2}}$ is t-score with area to the right of $\frac{\alpha}{2}$
 - s is the sample standard deviation _____
 - df is degrees of freedom of $n - 1$

! Important

Calculation with the t-distribution has to proceed with either a table or a calculator / computer - we'll cover what to do in SPSS

4 A Population Proportion

4.1 Introduction

! Important

I'm going to abridge this part a bit only because it gets a bit too 'in the weeds' for what we need to know

- Often, we are interested not just in the confidence interval of _____, but also of percentages and proportions
- First, we need to identify a situations that is more proportion-based than mean-based
 - In these scenarios, the underlying _____ is the binomial distribution

📢 Discuss: For review: write the notation formula to describe a binomial distribution and describe the characteristics of that distribution

4.2 Calculation of a Proportion CI

- Because of the underlying _____, we can take a proportion as: $P' = \frac{X}{N}$
 - Where:
 - X is a random variable of number of _____
 - n is the _____ of trials/sample size
 - P' is also sometimes shown as $\hat{P}$
- In the scenario that n is particularly _____ and p is not close to zero or one, we can actually treat this as normal like: $X \sim N(np, \sqrt{npq})$

- This all eventually works down to $\frac{X}{n} = P' \sim N\left(\frac{np}{n}, \frac{\sqrt{npq}}{n}\right)$
 - The second part after the comma is the treated as the “standard” error for
- One can calculate **error bound for a proportion (EBP)** to work towards a sort of confidence interval, but for proportions, just like we did for

5 Conclusion

5.1 A Plea for Confidence Intervals

- I am a big advocate of the value of confidence intervals in a lot of situations - I think they appropriately capture that these statistics really are just estimates/guesses!
- In my opinion, it is pretty much always appropriate to report confidence intervals alongside the point estimates of just about anything you calculate - it increases information and transparency

5.2 Recap

- Confidence intervals add valuable information about the relative spread and inaccuracy of sample statistics in means and in proportions
- Confidence intervals pretty much always become more narrow with larger sample size, continuing the trends demonstrated with the central limit theorem and the law of large numbers
- We introduced the t-distribution as a valuable way to calculate confidence intervals for means, even when lacking the population standard deviation

5.3 Lecture Check-in

- Make sure to complete and submit the lecture check-in

Key Terms

alpha The complement of the confidence level, effectively the percent of area that falls outside the interval estimate **6**

confidence interval The common form of an interval estimate used in inferential and traditional (frequentist) probabilistic statistics, suggests a range of values that a population parameter is likely to fall in given a certain level confidence 3

confidence level (CL) An arbitrarily set level of how certain one wants to be that the population parameter falls within a specified range 6

degrees of freedom (df) A measure of how many values can vary in a distribution's calculation (don't worry too much about this one) 9

error bound for a population mean (EBM) How far away from the point estimate which captures the middle some percent values determined by the confidence level 6

error bound for a proportion (EBP) Functions as sort of a 'appropriate standard deviation' for point estimates of proportions 12

inferential statistic A type of statistic meant to help infer characteristics of population or populations in relation to the sample statistics 3

interval estimate In frequentist statistics, sits as an inverse to a point estimate and gives a range as an estimate of the parameter 3

margin of error An alternative name for confidence intervals, used in polling often 4

point estimate A type of sample statistic that exists only as a single fixed value trying to represent the population parameter 2

reliability In this context, refers to how 'stable' the statistic is when taken many times 4

Student's t-distribution A distribution similar to the normal distribution but that also takes into account sample size, appropriate for use when the parameter standard deviation is not known 9

William Gosset Chemist and statistician at Guinness that came up with the t-distribution, wrote under the name of 'Student' 9